

Troubleshooting Guide: LACT Unit — Custody Transfer Metering Discrepancy

Category: Custody Transfer / Metering **Unit:** Lease Automatic Custody Transfer (LACT) Unit — Turbine or Positive Displacement Meter with Prover **Tools:** Meter proving data review, BS&W (basic sediment and water) analysis, meter factor trending **Fluid Package:** Not applicable in the VLE sense — custody transfer metering accuracy is governed by API MPMS standards and meter proving, not a phase-equilibrium calculation **Symptom:** A persistent volumetric discrepancy between LACT-metered delivered volume and receiving-party (pipeline/terminal) measured volume

1. Quick Reference

Item	Detail
Reported symptom	Persistent volumetric discrepancy between LACT unit delivered volume and the receiving party's measured volume, beyond normal tolerance
Initially unclear	Whether the meter itself had drifted out of calibration, or whether the crude being measured contained more free water/sediment than accounted for
Actual root cause	Rising basic sediment and water (BS&W) content in the crude, above the level the sampling/net-oil correction was capturing, meant a portion of measured "gross" volume was actually water rather than oil
Fix	Corrected the automatic sampler to capture a representative sample per API MPMS guidance; adjusted upstream dehydration/free-water knockout to reduce BS&W ahead of the LACT unit
Diagnostic signal	Meter factor from proving remained within normal tolerance (ruling out meter drift), while BS&W lab results trended up over the same period as the discrepancy
Prevention	Routine BS&W trending correlated against metering discrepancy; sampler system verification per API MPMS schedule

2. Symptom

- A **persistent volumetric discrepancy** developed between the LACT unit's metered delivered (net) volume and the volume measured by the receiving party (pipeline or terminal) — exceeding the normally accepted tolerance for custody transfer.

3. Why This Wasn't Assumed to Be Simple Meter Drift

A custody transfer discrepancy is often first suspected to be a **meter calibration/drift** issue, since that's the most direct explanation for "the numbers don't match." But custody transfer volume isn't just a raw meter reading — it's a **net oil volume**, calculated from gross metered volume corrected for BS&W (water and sediment) content via representative sampling. If the meter itself is accurate but the BS&W correction isn't representative of what's actually flowing, the reported "net oil" volume will be wrong even though the meter is working correctly.

4. Diagnostic Approach

Step 1 — Review meter proving history and meter factor trend

The LACT unit's **meter factor** (from routine proving against a certified prover) was reviewed over the relevant time period.

Finding: Meter factor remained **within normal tolerance** — ruling out meter mechanical drift/wear as the explanation for the discrepancy.

Step 2 — Review BS&W lab analysis trend

With the meter itself cleared, attention turned to the **net oil correction** — specifically, **BS&W (basic sediment and water) content** determined from the automatic sampler.

Finding: Lab BS&W results **trended upward** over the same period the volumetric discrepancy developed.

Step 3 — Assess sampler representativeness

The automatic sampler system was reviewed against **API MPMS (Manual of Petroleum Measurement Standards)** sampling guidance to confirm whether it was still collecting a truly representative sample of the flowing stream, particularly given the rising water content — free water tends to be harder to sample representatively than well-mixed oil, especially if flow conditions or sampler probe positioning aren't ideal for the current water cut.

Step 4 — Confirm the mechanism connecting BS&W and the discrepancy

With BS&W trending up and the sampler's representativeness in question, the mechanism became clear: **actual water content in the crude was higher than what the sampling system was capturing and correcting for**, meaning a portion of the volume being counted as "net oil" delivered was actually water — explaining the gap against the receiving party's independent measurement.

Quantitative Basis

- Meter factor across the last 4 provings: 0.9987–0.9993, all within the $\pm 0.25\%$ repeatability tolerance — confirms the meter itself was not drifting.
- Field water cut had risen from **2% to 9% over 3 months** (a nearby well coming online), directly increasing the BS&W load the sampler now had to capture representatively.
- Automatic sampler-reported BS&W: **0.3%**, essentially unchanged from its historical baseline. Independent manual lab draw taken concurrently: **1.1% actual BS&W** — a **0.8 percentage-point under-capture**.
- Custody transfer volumetric discrepancy grew from a baseline 0.3% (within the $\pm 0.5\%$ LACT tolerance) to **1.6%**, tracking the same 3-month period as the rising water cut.

5. Root Cause

Basic sediment and water (BS&W) content in the crude had risen, and the automatic sampling system was not capturing a fully representative sample of this water at the current flow/water-cut conditions. As a result, the **net oil volume correction understated the true water content**, causing the LACT unit to report a higher net oil volume than was actually being delivered — producing the discrepancy against the receiving party’s measurement.

6. Corrective Action

1. **Corrected the automatic sampler system** to capture a representative sample per API MPMS guidance (e.g., probe positioning, sample frequency, mixing considerations appropriate to the current water cut).
2. **Adjusted upstream dehydration/free-water knockout** ahead of the LACT unit to reduce BS&W entering the metering system in the first place.

7. Verification

- Post-correction, sampler-reported BS&W tracked independent manual lab draws to within **0.1 percentage points** (e.g., 1.0% sampler vs. 1.05% lab), versus the earlier 0.8-point gap.
- Volumetric discrepancy against the receiving party’s measurement dropped from 1.6% to **0.4%**, within the $\pm 0.5\%$ custody transfer tolerance.
- Held within tolerance across the **following 3 monthly ticket reconciliation cycles**.

8. Prevention / Long-Term Fix

- Established **routine BS&W trending correlated against metering discrepancy**, so a developing gap can be caught and investigated early rather than allowed to accumulate.
 - Added **sampler system verification per API MPMS schedule**, ensuring the sampling method remains representative as water cut and flow conditions change over time.
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9. General Troubleshooting Checklist (Reusable)

- ☐ Review meter proving history and meter factor trend first — rule in/out mechanical meter drift before looking elsewhere
- ☐ If the meter factor is stable/normal, review BS&W lab trend over the same period as the discrepancy
- ☐ Check automatic sampler representativeness against API MPMS guidance, especially if water cut has changed
- ☐ Confirm the mechanism: is the sampler under-capturing water content relative to what’s actually flowing?
- ☐ Correct sampler positioning/operation per API MPMS standards
- ☐ Address the source of rising BS&W upstream (dehydration, free-water knockout) where practical, not just the measurement of it
- ☐ Confirm resolution against the receiving party’s independent measurement, not just internal lab consistency
- ☐ Establish routine BS&W trending correlated to metering discrepancy as a standing check

10. Key Takeaway

A custody transfer volumetric discrepancy isn't automatically a meter calibration problem — check the meter factor from proving first, and if that's stable, look at the **net oil correction** itself. Rising BS&W content combined with a sampler that's no longer representative at the current water cut can make an accurate meter report an inaccurate net oil volume. Both the meter and the sampling/correction system need to be verified independently before concluding where a discrepancy is actually coming from.

Related Concepts / Tags

LACT-unit custody-transfer metering BS&W meter-factor meter-proving API-MPMS automatic-sampler net-oil-volume

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.